

Stats Chapter 1

Views

Frequentist view: Frequency or propensity of some phenomenon

Bayesian view: Reasonable expectation representing a state of knowledge or as quantification of personal belief

Definitions

Global: Collecting, organizing, summarizing, presenting & analyzing data as well as inferences, conclusions and decisions based on the analysis

Statistical: A statistic is a number derived from data (e.g. average, std)

Deterministic vs Probabilistic

Deterministic $\rightarrow$ Precisely determined ($F=ma$)

Probabilistic, Variability $\rightarrow$ Uncertainty, randomness $\rightarrow$ Statistical models

Terminology

- **Individuals** are the objects upon which we collect data. Individuals can be people, animals, plots of land and many other things.
- A **variable** is any characteristic of an individual.
- A **population** is a set of individuals that we are interested in studying.
- A **sample** is a subset of the individuals of a population.
- **Statistical Inference**: Attempting to draw conclusions about a population, based on observing a sample

Simple vs Stratified (Random sample)

Simple: size n consist of n individuals, each equally likely to be chosen $\rightarrow$ Reflects the population interested in studying

Stratified:

$\rightarrow$ Sampling units divide themselves into non overlapping groups that are homogeneous
strata

$\rightarrow$ Each random sample has separate (SRS)

$\rightarrow$ Combine to form sample space

Ensure each strata is not over/under represented

Collecting data

Experiments: Test set up w/ controlled conditions (manipulating interested variables, controlling uninterested ones)

Observational studies: Observing responses and uncontrolled inputs

Variable types

Categorical (Qualitative): Places individuals into groups/categories

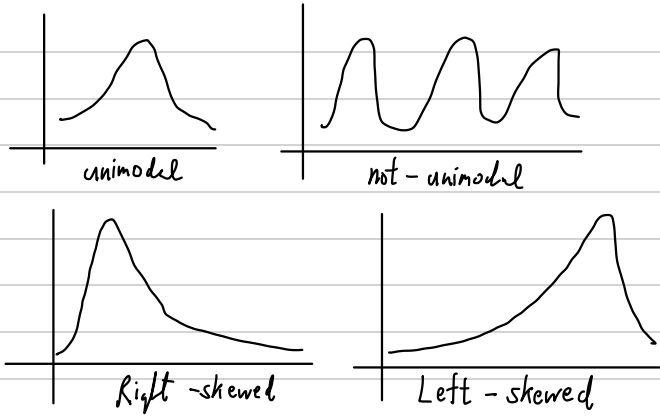
Quantitative: Numerical values, arithmetic operations are defined (Discrete or cont.)

Distribution of variable

↳ The frequency a variable takes on values

↳ Look for:

- overall pattern (shape, centre, spread)
- outliers (values that fall outside the pattern)
- Does it have 1 (unimodal) or more peaks
- Symmetric / Skewed



Trimmed mean

- Uses more info than median, resistant to outliers compared to mean
- 10% trimmed mean → discard top, lowest 10%

Spread

Range: Max-min, sensitive to extreme values, used to detect extremes

Variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

↳ degree of freedom, indep piece of info

Standard Deviation (std): $s = \sqrt{s^2}$

(Approximated as Range/4; it Range/3 for <10 terms; it Range/5 for >100 terms)

Plots

↳ Stemplot

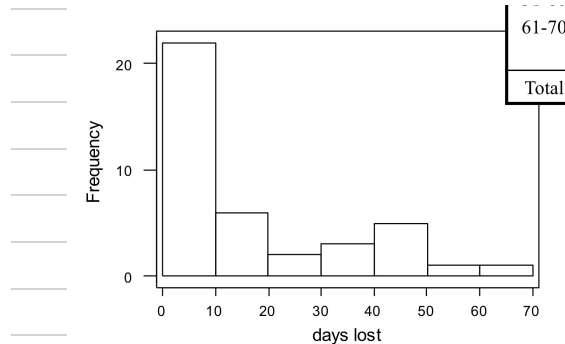
N = 50 (No. of terms)
Leaf Unit = 0.10

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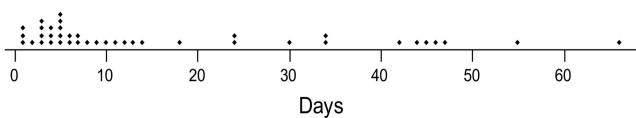
0 000111122233334444
0 55555566667889999
1 0011444
1 5669
2 03
2
3 1
3
4 2
    
```

↳ 0.9

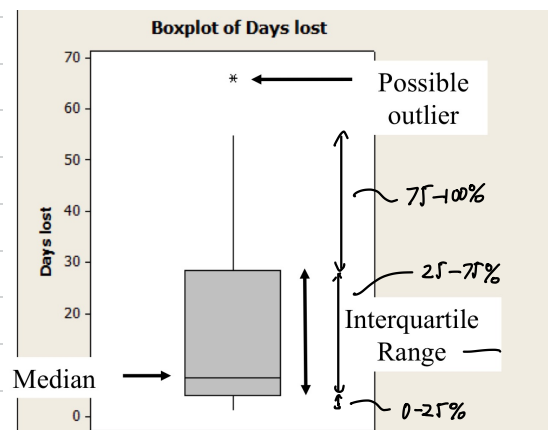
↳ Histogram



↳ Dot-plot



↳ Box-plot



Stats Chapter 2

Set theory

Definitions / Terminology

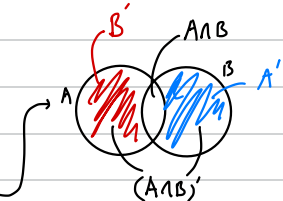
↳ Sample Space (S)
Event, Subset of S

↳ $A' =$ Complement of event A
 $A \cap B =$ Intersection (both occurring)
 $A \cup B =$ Union (At least one occurs)
 $\phi =$ Null event

↳ If $A \cap B = \phi \rightarrow A, B$ mutually exclusive

Theorems

$A \cap \phi = \phi$
 $A \cup \phi = A$
 $(A')' = A$
 $(A \cap B)' = A' \cup B'$



$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Probability

Definition

↳ $P(A) \rightarrow$ Probability of event A, the proportion of times the event will occur in infinitely many repetitions of the experiment

↳ $0 \leq P(A) \leq 1$

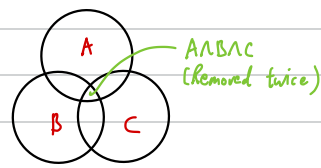
$$P(A) + P(A') = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(A \cap C) - P(B \cap C)$$

$$+ P(A \cap B \cap C)$$



↳ If $A_1, \dots, A_n$ mutually exclusive,
 $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

In addition, if $A_1 \cup A_2 \cup \dots \cup A_n = S$ then
 $\{A_1, \dots, A_n\}$ is a partition of S

↳ If # of outcome = k, equally likely to be selected,
each probability is $1/k$

Combination, Permutation

↳ nCr or $\binom{n}{r} = \frac{n!}{(n-r)!r!}$ (no ordering)

↳ $nPr = \frac{n!}{(n-r)!}$ (ordered)

↳ "n pick/choose r"

ex 4 suites pick 3

$$4C_3 = 4$$

$$4P_3 = 24$$

$$nPr \geq nCr$$

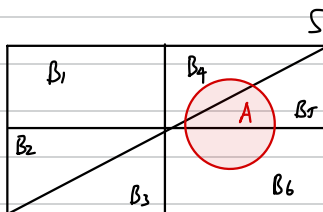
Conditional Prob.

↳ $P(B|A)$, prob. of B occurring given A occurred

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, P(A) > 0$$

↳ Then If events $B_1, \dots, B_k$ constitute a partition of S s.t. $P(B_i) \neq 0$ for $i=1, \dots, k$
Then any event A of S $P(A) = \sum_{i=1}^k P(B_i | A)$

Proof:



$$A = (A \cap B_1) \cup (A \cap B_2) \dots \cup (A \cap B_k)$$

$$P(A) = \sum_{i=1}^k P(B_i | A)$$

Mutual Independence

Definition: A collection of events $A_1, \dots, A_n$ are *mutually independent* iff for any subcollection $A_{i_1}, \dots, A_{i_k}$ we have

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j})$$

e.g. $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$

Theorem of total Probability

If events $B_1, \dots, B_k$ is a partition of S s.t. $P(B_i) > 0$ for $i = 1, \dots, k$, then for any event A of S :

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i) P(A|B_i)$$

Thm from cond. prob. Def. of cond. prob.

Bayes Rule

If events $B_1, \dots, B_k$ is a partition of S s.t. $P(B_i) > 0$ for $i = 1, \dots, k$, then for any event A in S s.t. $P(A) > 0$

$$P(B_n|A) = \frac{P(B_n \cap A)}{\sum_{i=1}^k P(B_i \cap A)} = \frac{P(B_n) P(A|B_n)}{\sum_{i=1}^k P(B_i) P(A|B_i)}, \text{ for } n=1, \dots, k$$

$P(A)$ def. of cond. prob. def. of cond. prob. def. of cond. prob. $P(A|B_n) = P(B_n \cap A) / P(B_n)$

Stats Chapter 3

Random variable

Def: A random variable (r.v.) is a function that associates a real number with each element in S

Discrete r.v.

Def: when its set of possible outcomes is countable (finite/infinite)

Probability mass function (pmf)

$f(x)$ of **discrete** r.v. $\bar{X}$ s.t. for each possible outcome $\bar{X}=x$

$$1.) P(\bar{X}=x) = f(x)$$

$$2.) f(x) \geq 0 \text{ \& } \sum_x f(x) = 1$$

Cumulative distribution function (cdf)

$$\hookrightarrow F(x) = P(\bar{X} \leq x) = \sum_{t \leq x} f(t), \text{ for } -\infty < x < \infty$$

for **discrete** r.v. $\bar{X}$ with pmf $f(x)$

$\hookrightarrow$ Properties:

- Monotonic $\longrightarrow F(x) \leq F(y)$
- Non-decreasing it $x \leq y$

$$\hookrightarrow P(a < \bar{X} \leq b) = F(b) - F(a)$$

Continuous r.v.

Def: If a sample space contains an ∞ # of possible outcomes, equal to the # of points of a line segment is called a cont. S

When a r.v. takes ∞ values on a cont. scale, it is a cont. r.v.

Probability of 0 of assuming exactly any values, hence we investigate intervals

Probability density function (pdf)

$f(x)$ of continuous r.v. $\bar{X}$, defined over set of Real numbers ($\mathbb{R}$) it:

$$1.) f(x) \geq 0 \text{ for all } x \in \mathbb{R}$$

$$2.) \int_{-\infty}^{\infty} f(x) dx = 1$$

$$3.) P(a < \bar{X} < b) = \int_a^b f(x) dx = P(a \leq \bar{X} \leq b) \\ = P(a \leq \bar{X} < b) \\ = P(a < \bar{X} \leq b)$$

Cumulative density function (cdf)

↳ $F(x)$ of cont. r.v. X with pdf $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \text{ for } -\infty < x < \infty$$

↳ Properties:

$$- P(a < X < b) = F(b) - F(a)$$

$$- f(x) = \frac{dF(x)}{dx}$$

Joint probability distributions

Joint pmt

Def: $f(x,y)$ is a joint pmt of **discrete** r.v.'s X and Y if

$$1.) f(x,y) \geq 0 \text{ for all } (x,y)$$

$$2.) \sum_x \sum_y f(x,y) = 1$$

$$3.) P(X=x, Y=y) = f(x,y)$$

Marginal pmt

$$f_X = g(x) = \sum_y f(x,y), \text{ for } X$$

$$f_Y = h(y) = \sum_x f(x,y), \text{ for } Y$$

↳ For any region A in x,y plane,
 $P[(X,Y) \in A] = \sum_A f(x,y)$

Joint pdf

Def: $f(x,y)$ is a joint pdf of **cont.** r.v.
 X and Y if

$$1.) f(x,y) \geq 0 \text{ for all } (x,y)$$

$$2.) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$3.) P[(X,Y) \in A] = \iint_A f(x,y) dA$$

Marginal pdf

$$f_X = g(x) = \int_{-\infty}^{\infty} f(x,y) dy, \text{ for } X$$

$$f_Y = h(y) = \int_{-\infty}^{\infty} f(x,y) dx, \text{ for } Y$$

Statistical independence

Def: X, Y are statistically independent iff.

$$f(x,y) = g(x)h(y) \text{ for all } (x,y)$$

Conditional distribution

Given $Y=y$:

$$f(x|y) = f(x,y) / h(y), h(y) > 0$$

Given $X=x$:

$$f(y|x) = f(x,y) / g(x), g(x) > 0$$

True for discrete, cont. r.v.

↳ $f(x,y) = g(x)$ as a result

Factorization

Factorizing $f(x,y) \rightarrow$ functions $a(x) b(y)$

is enough for independence, but the set of points where the joint pdf (or pmt) is +ve must be the cartesian product of the set of points where marginals are +ve

$$f(x,y) = a(x) * b(y) * \underset{\substack{\uparrow \\ \text{const.}}}{c}$$

Statistical Mutual Independence

Def: Let $\bar{X}_1, \dots, \bar{X}_n$ be n r.v.'s (discrete or cont.), with joint pdf (or pmf) $f(x_1, \dots, x_n)$ and marginals $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$.
 $\bar{X}_1, \dots, \bar{X}_n$ are statistically mutually indep. iff

$$f(x_1, x_2, \dots, x_n) = f_1(x_1) f_2(x_2) \dots f_n(x_n)$$

Stats Chapter 4

Expectation (expected value or mean)

For r.v. $\bar{X}$, the expected average value $\bar{X}$ is

$$\mu = E(\bar{X}) = \begin{cases} \sum_x x f(x), & \text{disc. } \bar{X} \\ \int_{-\infty}^{\infty} x f(x), & \text{cont. } \bar{X} \end{cases}$$

For function $g(\bar{X})$, function of $\bar{X}$

$$E[g(\bar{X})] = \begin{cases} \sum_x g(x) f(x), & \text{disc. } \bar{X} \\ \int_{-\infty}^{\infty} g(x) f(x), & \text{cont. } \bar{X} \end{cases}$$

Properties of Expectation

1.) $E(a\bar{X} + b) = aE(\bar{X}) + b$

2.) $E(a) = a$

3.) $E[g(\bar{X}) + h(\bar{X})] = E[g(\bar{X})] + E[h(\bar{X})]$

4.) For non-negative r.v. $\bar{X}$

a.) $E(\bar{X}) \geq 0$

b.) If $E(\bar{X}) = 0 \iff P(\bar{X} = 0) = 1$

Variance

$\hookrightarrow \sigma^2$, measure of how closely concentrated the dist is around μ

$$\sigma^2 = E[(\bar{X} - \mu)^2] = E(\bar{X}^2) - \mu^2$$

$$\sigma^2 = \begin{cases} \sum_x (x - \mu)^2 f(x) \\ \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \end{cases}$$

$\hookrightarrow \sigma_{g(\bar{X})}^2$, variance of function

$$= E[(g(\bar{X}) - \mu_{g(\bar{X})})^2] = E[g^2(\bar{X})] - E[g(\bar{X})]^2$$

$$= \sum_x [g(x) - \mu_{g(\bar{X})}]^2 f(x)$$

$$= \int_{-\infty}^{\infty} [g(x) - \mu_{g(\bar{X})}]^2 f(x) dx$$

Properties of variance

$\bar{X}, Y$ r.v. and a, b const.

1.) $\text{Var}(a\bar{X} + b) = a^2 \text{Var}(\bar{X})$

2.) $\text{Var}(\bar{X}) \geq 0$

3.) $\text{Var}(\bar{X}) = 0 \iff \bar{X} = E(\bar{X})$ w/ prob 1

4.) $\text{Var}(a) = 0$

Joint dist, expectation & variance

$$E[g(X, Y) + h(X, Y)] = E[g(X, Y)] + E[h(X, Y)]$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \underbrace{E[(X - E(X))(Y - E(Y))]}_{\text{Covariance}}$$

Covariance

- ↳ Measurement of the nature of the association b/w 2 r.v.s
- ↳ Only describes linear relation b/w X, Y

$$\sigma_{XY}^2 = \text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))] = E[XY] - E(X)E(Y)$$

$$\begin{cases} \text{Large } \bar{X}, \text{ Large } Y \longrightarrow +ve \text{ Cov} \\ \text{Small } \bar{X}, \text{ Small } Y \end{cases} \quad \begin{cases} \text{Small } \bar{X}, \text{ Large } Y \longrightarrow -ve \text{ Cov} \\ \text{Small } \bar{X}, \text{ Large } Y \end{cases}$$

Properties of Cov.

- 1.) $\text{Cov}(X, X) = \text{Var}(X)$
- 2.) $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- 3.) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$
- 4.) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

Independence

- 1.) If X, Y indep, $E(XY) = E(X)E(Y)$, $\text{Cov}(X, Y) = 0$ (Vice Versa n.t necessarily true)
- 2.) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Correlation coefficient

- ↳ Measure of strength and direction of the linear relationship b/w X, Y

↳

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$= \frac{E(XY) - E(X)E(Y)}{\sqrt{(E(X^2) - E^2(X))(E(Y^2) - E^2(Y))}}$$

- ↳ Scale-free: $\rho(aX + b, cY + d) = \rho(X, Y)$

- ↳ $-1 \leq \rho(X, Y) \leq 1$

Stats Chapter 5

Uniform discrete

For r.v. assuming values $x_1, \dots, x_k$ with equal prob.

$$\hookrightarrow f(x; k) = 1/k, x = x_1, x_2, \dots, x_k$$

$$\hookrightarrow E(X) = \mu = \frac{1}{k} \sum_{i=1}^k x_i$$

$$\hookrightarrow \text{Var}(X) = \frac{1}{k} \sum_{i=1}^k (x_i - \mu)^2$$

Bernouli trial

$\hookrightarrow$ R.v. only has two outcomes: success / failure

Bernouli Distribution

$$X = \begin{cases} 1, \text{ success} \\ 0, \text{ failure} \end{cases} \quad \begin{array}{c|c|c} x & 0 & 1 \\ \hline f(x) & q & p \end{array}$$

$$\hookrightarrow \mu = p$$

$$\hookrightarrow \sigma^2 = pq$$

Binomial Dist

$\hookrightarrow$ # of success in n indep. Bernouli trials

$$b(x; n, p) = f(x) = \binom{n}{x} p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

$$\hookrightarrow \mu = np$$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\hookrightarrow \sigma^2 = npq$$

Multinomial dist.

$\hookrightarrow$ Trials w/ more than 2 possible outcomes

$\hookrightarrow$ For trials that can result in k outcomes $E_1, E_2, \dots, E_k$

with probabilities $p_1, p_2, \dots, p_k$, then the prob. dist. of r.v.s

$X_1, X_2, \dots, X_k \rightarrow$ number of occurrences for $E_1, E_2, \dots, E_k$ in n indep. trials is

$$f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k, n) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}, \text{ where } \sum_{i=1}^k x_i = n, \sum_{i=1}^k p_i = 1$$

Hypergeometric distribution

↳ Sampling w/o replacement (dependent trials)

↳ prob of hypergeometric r.v. X , the # of successes in a random sample of size n selected from N items of which k are labelled success and $N-k$ are labelled failure is:

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \max\{0, n-(N-k)\} \leq x \leq \min\{n, k\}$$

$$\rightarrow \mu = \frac{nk}{N}$$

$$\rightarrow \sigma^2 = \frac{N-n}{N-1} \cdot \frac{n}{N} \cdot k \left(1 - \frac{k}{N}\right)$$

Acceptance Sampling

↳ Sampling where some quantity is selected and tested

Hypergeometric & Binomial

↳ For $n \ll N$, $p \sim k/N$
used to approx. hypergeometric dist.

↳ Rule of thumb $\frac{n}{N} \ll 0.05$

Multivar. hypergeometric

$$f(x_1, x_2, \dots, x_k; a_1, a_2, \dots, a_k, N, n) = \frac{\binom{a_1}{x_1} \binom{a_2}{x_2} \dots \binom{a_k}{x_k}}{\binom{N}{n}} \rightarrow N \text{ items, partitioned into } a_1, a_2, \dots, a_k \text{ selecting } n$$

$$\rightarrow \sum_{i=1}^k x_i = n, \sum_{i=1}^k a_i = N$$

Negative Binomial

↳ Prob of k^{th} success occurring on the x^{th} trial in indep. Bernoulli trials

$$b^*(x; k, p) = f(x) = \binom{x-1}{k-1} p^k q^{x-k}, x = k, k+1, \dots$$

$$\rightarrow \mu = k/p$$

$$\rightarrow \sigma^2 = \frac{k(1-p)}{p^2}$$

Geometric Dist.

↳ X : number of trial where the first success occurs in indep. Bernoulli Trials.

$$g(x; p) = f(x) = p q^{x-1}, x = 1, 2, 3, \dots$$

special case of neg binomial where $k=1$

$$\rightarrow \mu = 1/p$$

$$\rightarrow \text{Var} = \frac{1-p}{p^2}$$

Memoryless Property

↳ Given X follows geometric dist. for integers $s > t$
Then,

$$P(X > s | X > t) = P(X > s-t)$$

↳ Intuition: Prob. of getting an additional $s-t$ set of failures, if already observed t failures is the same as prob. of observing $s-t$ failures at the start

$$\hookrightarrow P(X > s | X > t) = \frac{P(X > s \text{ and } X > t)}{P(X > t)} = \frac{P(X > s)}{P(X > t)} = \frac{(1-p)^s}{(1-p)^t} = (1-p)^{s-t} = P(X > s-t)$$

Poisson Process

- 1.) Number of outcomes occurring in 1 time interval / specified region is indep. of the number that occur in any other disjoint time interval / region. (No memory)
- 2.) Prob. that a single outcome will occur during a very short time interval / in a small region is proportional to the length of the time interval / the size of the region, **does not** depend on number of outcomes occurring outside this time interval / region
- 3.) Prob. that more than 1 outcome will occur in such a short time interval / fall in such a small region is negligible.

Poisson Distribution

$$p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, x = 0, 1, 2, \dots$$

$\lambda \rightarrow$ avg. number of outcome per unit time, distance, area, vol. etc

$t \rightarrow$ specified the interval / region

$$\hookrightarrow \mu = \sigma^2 = \lambda t$$

Approx. of binomial by Poisson dist.

As $n \rightarrow \infty$, $p \rightarrow 0$, and $np \xrightarrow{n \rightarrow \infty} \mu$ remains constant,

$$b(x; n, p) \xrightarrow{n \rightarrow \infty} p(x; \mu)$$

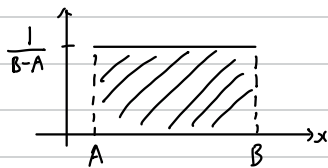
↳ Approx. binomial prob. for "rare occurrences"

↳ Rule of thumb: 1.) $n \geq 20$ & $p \leq 0.05$
 $n \geq 100$ & $np \leq 10$

Stats Chapter 6

Uniform Continuous

$$f(x; A, B) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B \\ 0, & \text{o/w} \end{cases}$$



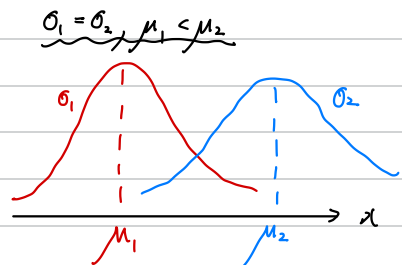
$$\hookrightarrow \mu = A+B/2$$

$$\hookrightarrow \sigma^2 = (B-A)^2/12$$

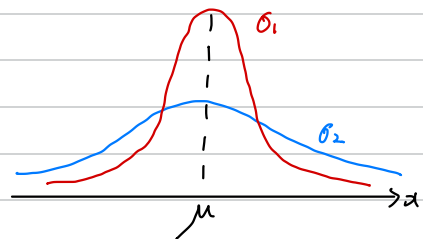
Normal Distribution

pdf of normal r.v. $\bar{X}$ w/ mean μ and variance σ^2 is

$$n(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty$$



$\sigma_2 > \sigma_1, \mu_1 = \mu_2$



Standardization & z-scores

$\hookrightarrow$ Transform any normal r.v. $\bar{X} \longrightarrow$ r.v. Z with $\mu=0, \sigma=1$

$$\hookrightarrow Z = (\bar{X} - \mu) / \sigma \longrightarrow Z \sim N(0, 1)$$

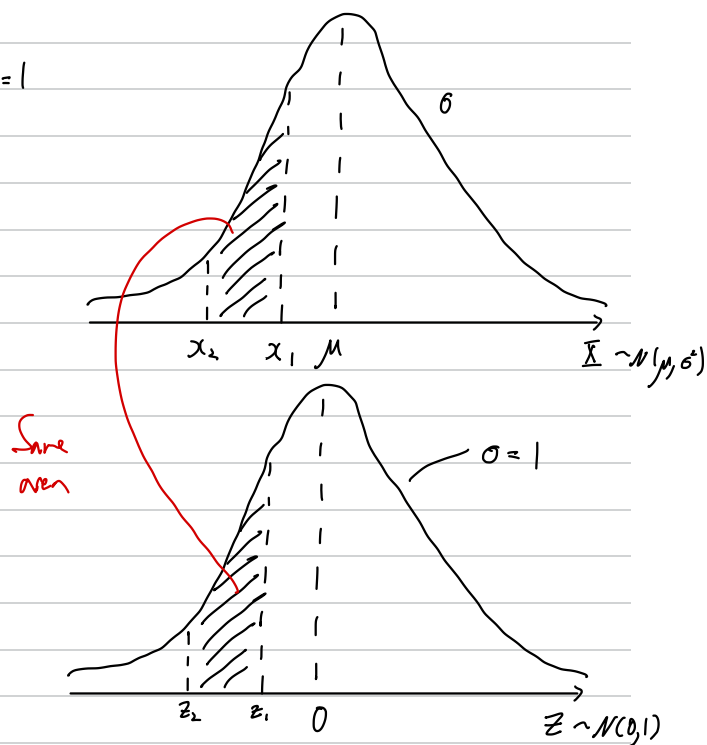
$$\begin{aligned} \hookrightarrow P(x_1 < \bar{X} < x_2) &= \int_{x_1}^{x_2} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} dx \\ &= \int_{z_1}^{z_2} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz = P(z_1 < Z < z_2) \end{aligned}$$

$\hookrightarrow$ z-value or z-score tells us how many std, the original observation falls away from the mean $\longrightarrow x = \mu + \sigma z$

$$z = \frac{x - \mu}{\sigma}$$

Percentile

$$P(Z < z) = p, \quad P(\bar{X} < x) = p \quad \text{for r.v. } \bar{X}$$



Normal approximation to binomial

↳ If X is binomial w/

1.) Mean $\mu = np$

2.) Variance $\sigma^2 = npq$

Then limiting form $Z = \frac{\bar{X} - np}{\sqrt{npq}}$ as $n \rightarrow \infty$ is standard normal

↳ Accurate for n large, p not extremely close to 0, 1

↳ Okay when n small, $p \sim 0.5$

$$\hookrightarrow P(x_1 \leq \bar{X} \leq x_2) = \sum_{k=x_1}^{x_2} b(k; n, p) \approx P\left(\frac{x_1 - 0.5 - np}{\sqrt{npq}} < Z < \frac{x_2 + 0.5 - np}{\sqrt{npq}}\right)$$

continuity correction

↳ Rule of thumb: If $np \geq 5$ and $nq \geq 5$

Gamma Function

$$\Gamma(x) = \int_0^{\infty} x^{x-1} e^{-x} dx, \text{ for } x > 0$$

a.) $\Gamma(n) = \frac{(n-1)(n-2)\dots 1 \Gamma(1)}{(n-1)!}$; find using by parts

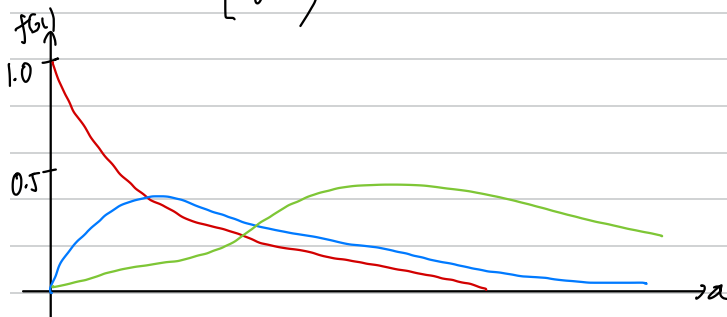
b.) $\Gamma(a) = (a-1) \Gamma(a-1)$

c.) $\Gamma(1) = 1$

d.) $\Gamma(1/2) = \sqrt{\pi}$

Gamma dist.

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0 \\ 0, & \text{elsewhere} \end{cases} \quad \alpha, \beta > 0$$



↳ $\mu = \alpha\beta$

↳ $\sigma^2 = \alpha\beta^2$

Exponential dist.

↳ Special case of gamma dist. when $\alpha=1$

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}, \beta > 0$$

↳ $\mu = \beta$

↳ $\sigma^2 = \beta^2$

Poisson Process and exponential dist.

↳ λ = mean # of events per unit time

① Probability of no events leading up to time t in a Poisson process,

$$p(0|t) = \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t}$$

② $P(X > t) = e^{-\lambda t}$, let X be the 1st Poisson event

③ Cumulative distribution function, $P(0 \leq X \leq x) = 1 - e^{-\lambda x}$

④ Differentiate — Density function, $f(x) = \lambda e^{-\lambda x}$, $\lambda = \frac{1}{\beta}$  β : mean time b/w failures

Memoryless Property of exponential dist.

$$P(X \geq t) = P(X \geq t_0 + t | X \geq t_0)$$

If a component "make it" to t_0 hours, the prob of it lasting an additional t hours is equal to it lasting t hours from the beginning

Incomplete Gamma Function

$$F(x; \alpha) = \int_0^x \frac{y^{\alpha-1} e^{-y}}{\Gamma(\alpha)} dy$$

ex. $F(6; 5) = \int_0^6 \frac{y^{5-1} e^{-y}}{\Gamma(5)} dy$ (usually in appendix)

Chi-Squared dist

↳ Special case of exponential dist, $\alpha = \nu/2$; $\beta = 2$

↳ $\nu \in \mathbb{Z}^+$, degrees of freedom

$$f(x; \nu) = \begin{cases} \frac{x^{\nu/2-1} e^{-x/2}}{2^{\nu/2} \Gamma(\nu/2)}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

↳ $\mu = \nu$

↳ $\sigma^2 = 2\nu$

Weibull dist

$$\hookrightarrow f(x; \alpha, \beta) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}, \alpha, \beta > 0$$

$$\hookrightarrow F(x) = 1 - e^{-\alpha x^\beta}$$

$$\hookrightarrow \mu = \alpha^{-1/\beta} \Gamma\left(1 + \frac{1}{\beta}\right)$$

$$\hookrightarrow \sigma^2 = \alpha^{-2/\beta} \left\{ \Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right) \right]^2 \right\}$$

Failure rate

$\hookrightarrow$ Reliability as the probability that it will function properly for at least a specified time under specified experimental conditions.

$$R(t) = P(T > t) = \int_t^\infty f(t) dt = 1 - F(t)$$

$\hookrightarrow$ Failure rate,

$$Z(t) = \lim_{\Delta t \rightarrow 0} \frac{F(t+\Delta t) - F(t)}{\Delta t} \cdot \frac{1}{R(t)} = \frac{f(t)}{R(t)} = \frac{f(t)}{1 - F(t)}$$

$$= \alpha \beta t^{\beta-1}, t > 0$$

$\hookrightarrow$ Conditional prob. that a component last an additional Δt given it lasted time t

a) If $\beta=1$, failure rate = α (Lack of memory, exponential dist)

b) $\beta > 1$, $Z(t)$ is an increasing function, it wears over time

c) $\beta < 1$, $Z(t)$ is a decreasing function, it strengthens over time.

Stats Chapter 7

Transformation of discrete r.v.

Given:

$\hookrightarrow X$ is a discrete r.v.

$\hookrightarrow$ pmf of $X = f(x)$

$\hookrightarrow Y = u(X)$, one to one
 $\hookrightarrow y = u(x) \Leftrightarrow x = w(y)$ solving

Then the pmf of Y :

$$\hookrightarrow g(y) = f[w(y)]$$

Transformation of multiple discrete r.v.

Given:

$\hookrightarrow X_1, X_2$ discrete r.v.

$\hookrightarrow$ joint pmf $f(x_1, x_2)$

$\hookrightarrow Y_1 = u_1(X_1, X_2), Y_2 = u_2(X_1, X_2)$, one-to-one

$$x_1 = w_1(y_1, y_2), x_2 = w_2(y_1, y_2)$$

Then joint pmf of Y_1, Y_2 :

$$\hookrightarrow g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)]$$

Transformation of cont. r.v.

Given:

↳ X is a **cont.** r.v.

↳ pdf of $X \rightarrow f(x)$

↳ $Y = u(X)$, one to one

$$\hookrightarrow y = u(x) \Leftrightarrow x = w(y)$$

Then, pdf of Y :

$$\hookrightarrow g(y) = f[w(y)] \cdot |J|$$

where $J = w'(y)$ is the Jacobian of the transformation

Transformation of multiple cont r.v.

Given:

↳ X_1, X_2 are **cont** r.v.

↳ joint pdf of $f(x_1, x_2)$

↳ $Y_1 = u_1(X_1, X_2), Y_2 = u_2(X_1, X_2)$

$$\hookrightarrow y_1 = u_1(x_1, x_2), y_2 = u_2(x_1, x_2) \Leftrightarrow x_1 = w_1(y_1, y_2), x_2 = w_2(y_1, y_2)$$

Then, joint pdf of Y_1, Y_2 :

$$\hookrightarrow g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)] \cdot |J|$$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Transformation of non-1-to-1 r.v.

Given:

↳ X is a **cont.** r.v.

↳ pdf $\rightarrow f(x)$

↳ $Y = u(X)$, not 1-to-1

If X can be partitioned into k mutually disjoint sets, s.t. each inverse function,

$$x_1 = w_1(y), x_2 = w_2(y) \dots x_k = w_k(y)$$

of $y = u(x)$ defines a 1-to-1 function, the pdf of Y is

$$g(y) = \sum_{i=1}^k f[w_i(y)] |J_i|$$

$$J_i = w_i'(y), i=1, 2, \dots, k$$

Moments

↳ r^{th} moment about origin of r.v. X is:

$$\mu_r' = E(X^r) = \begin{cases} \sum_x x^r f(x) & , X \text{ disc.} \\ \int_{-\infty}^{\infty} x^r f(x) dx & , X \text{ cont.} \end{cases}$$

↳ Mean, $\mu = \mu_1'$

↳ Variance, $\sigma^2 = \mu_2' - (\mu_1')^2$

$$\hookrightarrow \mu_r' = \left. \frac{d^r M_X(t)}{dt^r} \right|_{t=0}$$

Moment-generating func.

$$M_X(t) = E(e^{tx}) = \begin{cases} \sum_x e^{tx} f(x) & , X \text{ disc.} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & , X \text{ cont.} \end{cases}$$

Central moment

↳ r^{th} moment around μ

$$E[(X-\mu)^r]$$

$r=1$	$r=2$	$r=3$	$r=4$
$E[(X-\mu)^1] = 0$	σ^2	skewness	kurtosis

Uniqueness

↳ Let X, Y be 2 r.v., $M_X(t), M_Y(t)$ are two moment-gen. func.

If $M_X(t) = M_Y(t)$ for all values of t ,

then X, Y have same prob. dist.

Properties of moment-gen func.

$$\hookrightarrow M_{X+a}(t) = e^{at} M_X(t)$$

$$\hookrightarrow M_{aX}(t) = M_X(at)$$

$\hookrightarrow$ If $X_1, X_2, \dots, X_n$ are r.v. with $M_{X_1}(t), M_{X_2}(t), \dots, M_{X_n}(t)$

$$Y = X_1 + X_2 + \dots + X_n$$

$$\text{Then } M_Y(t) = M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t)$$

Linear comb. of r.v.

Let $X_1, X_2, \dots, X_n$ be r.v.s with normal dist.

$\mu_1, \mu_2, \dots, \mu_n$ be their means

$\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$ be their variances

$$Y = a_1 X_1 + a_2 X_2 + \dots + a_n X_n$$

Then,

$$\mu_Y = a_1 \mu_1 + a_2 \mu_2 + \dots + a_n \mu_n$$

$$\sigma_Y^2 = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \dots + a_n^2 \sigma_n^2$$

For chi-squared dist.

$\hookrightarrow$ Let $X_1, X_2, \dots, X_n$ be mutually indep. r.v. in chi-squared dist. with $v_1, v_2, \dots, v_n$ degrees of freedom,

The r.v. $Y = X_1 + X_2 + \dots + X_n$

has chi-squared dist with $\sum_{i=1}^n v_i$ deg. of freedom

Chi-squared and Normal dist.

Let $X_1, X_2, \dots, X_n$ are indep. r.v., with **identical** normal dist,
mean, μ
variance, σ^2

$$\text{Then r.v. } Y = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2$$

has chi-squared dist, $v=n$ deg. of free dom

Let $X_1, X_2, \dots, X_n$ are indep. r.v., with normal dist
mean, μ_i
variance, σ_i^2 for $i=1, 2, \dots, n$

$$\text{Then r.v. } Y = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$$

has chi-squared dist, $v=n$ deg. of free dom

Stats Chapter 8

Definitions

- ↳ Population: Totality of observations where we are concerned
- ↳ Sample: Subset of population
- ↳ Parameter: A number that describes the population, fixed but in practice unknown
- ↳ Statistic: Function of sample data, r.v. w/ distribution

independent, identically distributed

Let $X_1, X_2, \dots, X_n$ be n indep. r.v., each having the same proba. dist. $f(x)$

Define $X_1, X_2, \dots, X_n$ to be a random sample of size n from population $f(x)$, the joint pdf is

$$f(x_1, x_2, \dots, x_n) = f(x_1) f(x_2) \dots f(x_n)$$

↳ Sampling dist: The proba. dist. of a statistic

Depends on:

- Size of population
- Size of sample
- Method of sampling

Sampling from Normal

↳ r.s. of n taken from $N(\mu, \sigma^2)$,

↳ each observation $\bar{X}_j$, $j=1, \dots, n$ have same dist. $\rightarrow \bar{X}_1, \bar{X}_2, \dots, \bar{X}_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Then $\bar{\bar{X}} \sim N(\mu, \frac{\sigma^2}{n})$, where $\bar{\bar{X}} = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$

Central Limit Theorem

If $\bar{X}$ is the mean of r.s. of sample size n from a population w/

↳ Mean, μ

↳ Variance, σ^2

Then the limiting form of distribution of $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ as $n \rightarrow \infty$ is standard normal, $N(z; 0, 1)$

standard error, σ_z

↳ Good for $n \geq 30$

Inferences on Population Mean

① Given real average $\bar{x}$, std (standard deviation) σ , conjectured mean μ , sample size n

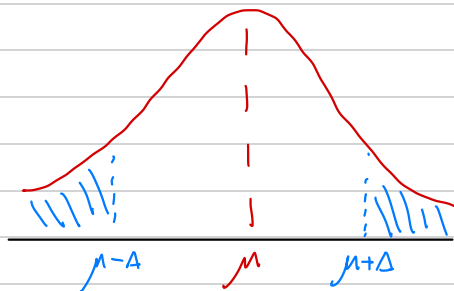
② Find $P(|\bar{X} - \mu| \geq \Delta)$

$$= P(\bar{X} - \mu \geq \mu + \Delta) + P(\bar{X} - \mu \leq \mu - \Delta)$$

$$\Delta = |\bar{x} - \mu|$$

$$= 2P\left[\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq 100(\bar{x} - \mu)\right]$$

$$= 2P(Z \geq 100(\bar{x} - \mu))$$



③ Convert on result

Sample Dist. of difference b/w 2 means $(\mu_1 - \mu_2)$

Given 2 populations with

↳ Means, μ_1, μ_2

↳ Variances, σ_1^2, σ_2^2

↳ Sample size n_1, n_2 respectively

Then $\bar{X}_1 - \bar{X}_2$ is approximately normally distributed w/

↳ Mean, $\mu_1 - \mu_2$

↳ Variance, $\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$

By Central Limit Theorem,

$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$ is approximately standard normal

Law of Large numbers

Let $\bar{X}_1, \bar{X}_2, \dots$ iid w/ μ, σ^2

If $\bar{X}_n = \sum_{i=1}^n \bar{X}_i$ and $n \rightarrow \infty$

Then $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \rightarrow 0$,
 $\bar{X}_n \rightarrow E(\bar{X}_n) = \mu$

Weak Law of Large numbers

Let $\bar{X}_1, \bar{X}_2, \dots$ iid w/ $\mu, \sigma^2 < \infty$

Then for all $\epsilon > 0$,

$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| < \epsilon) = 1$ or

$\bar{X}_n$ converges to μ in probability

Strong Law of Large numbers

$P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1$

Almost sure convergence

Chebyshev's Inequality

The probability that any r.v. within k stds of the mean is:

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

Markov's inequality

If X is a r.v. ≥ 0 , $E(X) < \infty$ and $a > 0$

Then: $P(X \geq a) \leq \frac{E(X)}{a}$

Sampling dist. of S^2

If S^2 is the variance of a r.s. of size n , taken from a normal population with variance σ^2 ,
Then:

$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \frac{(\bar{X}_i - \bar{X})^2}{\sigma^2}$ has a reduced chi-squared dist, d.o.f, $\nu = n-1$

↳ Each $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$

T Distribution

Let Z be a standard normal variable,
 V be a χ^2 r.v., $\bar{v}$ d.o.f.

If Z, V indep., then

$$T = \frac{Z}{\sqrt{V/\bar{v}}} \text{ has density function } \rightarrow h(t) = \frac{\Gamma[(\bar{v}+1)/2]}{\Gamma(\bar{v}/2) \sqrt{\pi \bar{v}}} \left(1 + \frac{t^2}{\bar{v}}\right)^{-(\bar{v}+1)/2}, \quad -\infty < t < \infty$$

t distribution $\rightarrow \bar{v}$ d.o.f.

Corollary

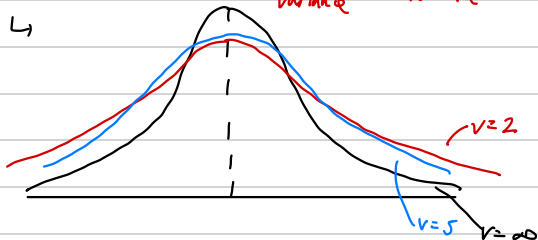
Let $X_1, X_2, \dots, X_n$ be indep. r.v., normal with mean μ , std σ ,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Then the r.v. $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ has t -dist., $\bar{v} = n-1$ d.o.f.

$\hookrightarrow$ If n is small, S fluctuates from sample groups significantly
 T has more variability compared to Z

$\hookrightarrow n \geq 30$, then $\frac{S^2}{\sigma^2} \sim \chi^2$
sample variance population variance



$\hookrightarrow t_{\alpha} = \alpha$ value where right of it
 $\uparrow$ Probability has area α
value

$$\hookrightarrow t_{1-\alpha} = -t_{\alpha}$$

F-Distribution

Let U, V be 2 indep. r.v. w/ chi-squared dist., v_1, v_2 d.o.f.

The dist. of the r.v.

$$F = \frac{U/v_1}{V/v_2} \quad \begin{array}{l} \text{F-distribution} \\ \text{d.o.f } v_1, v_2 \end{array}, \quad h(f) = \begin{cases} \frac{\Gamma[(v_1+v_2)/2] \cdot (v_1/v_2)^{v_1/2}}{\Gamma(v_1/2) \cdot \Gamma(v_2/2)} \cdot \frac{f^{(v_1/2)-1}}{(1 + v_1 f/v_2)^{(v_1+v_2)/2}}, & f > 0 \\ 0, & f \leq 0 \end{cases}$$

$$\hookrightarrow f_{1-\alpha}(v_1, v_2) = \frac{1}{f_{\alpha}(v_2, v_1)}$$

Corollary

If S_1^2 and S_2^2 are variances of indep. r.v.s. of size n_1, n_2 w/ normal populations w/ variances σ_1^2, σ_2^2
Then

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_1^2} \text{ has } F\text{-dist.}, \text{ d.o.f. of } n_1-1, n_2-1$$

Stats Chapter 9

Point Estimate

↳ A point estimate of some population parameter, θ is a single value $\hat{\theta}$ of a statistic $\hat{\theta}$
 $\text{eg } \mu \text{ or } p \quad \downarrow \quad \downarrow$
 $\quad \quad \quad \bar{x}$

Unbiased estimator

↳ A statistic $\hat{\theta}$ is said to be an unbiased estimator of the parameter if

$$E(\hat{\theta}) = \theta$$

↳ If $\bar{X}$ is an unbiased estimator of μ , $E(\bar{X}) = \mu$

$$E(S^2) = \sigma^2$$

Variance of a point estimator

↳ Most efficient estimator → The unbiased estimator of parameter θ that has the **smallest** variance

↳ Standard deviation of statistic → **std error** = $\sigma / \sqrt{n}$
(or estimator)

Interval estimation

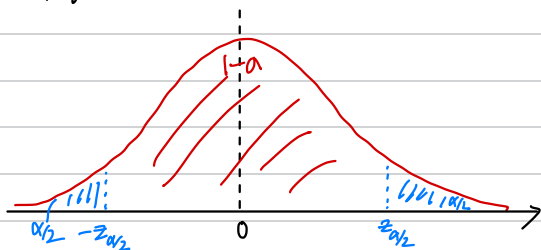
↳ Confidence Interval (CI) reports a large range of possible θ values,
measures the degree of reliability of a confidence interval

$$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha = 100\%(1 - \alpha), \quad 0 < \alpha < 1, \text{ Confidence level}$$

$$\hat{\theta}_L < \theta < \hat{\theta}_U, \text{ Confidence interval}$$

↳ Want **narrow interval**, **high confidence level**

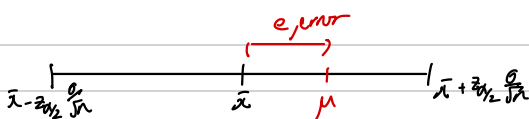
Sampling from normal



$z_{\alpha/2} \rightarrow z$ that gives area of $\alpha/2$

↳ Confidence interval for $\mu \rightarrow -z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} < z_{\alpha/2} \rightarrow \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad P(\dots) = 1 - \alpha$
Confidence level

↳ Error, If $\bar{x}$ is used as an estimate of μ , we can be 100%(1- α)% confident that the error, $|\mu - \bar{x}|$ will not exceed $z_{\alpha/2} \sigma / \sqrt{n}$



↳ knowing a sample size of ($\star$ Round up!)

$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{e} \right)^2$ is needed to ensure a $100(1-\alpha)\%$ confident that error is less than a number e

One-sided bounds

Upper

CI: $\mu < \bar{X} + z_{\alpha} \sigma / \sqrt{n}$
bound, $\bar{x} + z_{\alpha} \sigma / \sqrt{n}$

Lower

CI: $\mu > \bar{X} - z_{\alpha} \sigma / \sqrt{n}$
bound, $\bar{x} - z_{\alpha} \sigma / \sqrt{n}$

CI for μ , Unknown σ (T-dist)

If $\bar{x}$, s , n are std deviation of r.v. from normal popltn with unknown variance σ^2 ,
the $100(1-\alpha)\%$ CI for μ is:

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}, \quad t_{\alpha/2} \text{ is the t-value with dof } r=n-1, \text{ area } \alpha/2$$

Prediction Interval

↳ Interval for a single new obs X_0

$\bar{X} \sim N(\mu, \sigma^2/n) \xleftrightarrow{\text{independent}} X_0 \sim N(\mu, \sigma^2)$

↳ Define $z = \frac{x_0 - \bar{x}}{\sqrt{\sigma^2 + \sigma^2/n}}$

Predictor interval

$$\bar{x} - z_{\alpha/2} \sigma \sqrt{1 + 1/n} < x_0 < \bar{x} + z_{\alpha/2} \sigma \sqrt{1 + 1/n}$$

or

$$\bar{x} - t_{\alpha/2} s \sqrt{1 + 1/n} < x_0 < \bar{x} + t_{\alpha/2} s \sqrt{1 + 1/n}$$

Two samples

① Variance known

$$CI: (\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

② Variance unknown, equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$)

$$\hat{\sigma}^2 = \underbrace{S_p^2}_{\text{pooled estimate}} = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

$$CI: (\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + \underbrace{t_{\alpha/2}}_{\text{w/ d.o.f } n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

③ Unknown, unequal variances

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad \text{w/ d.o.f. } \gamma = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{\left[(S_1^2/n_1)/(n_1-1) + (S_2^2/n_2)/(n_2-1) \right]}, \quad \star \text{ Round down!}$$

$$CI: (\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + \underbrace{t_{\alpha/2}}_{\text{w/ d.o.f. } \gamma} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Paired obs

↳ Doing 2 obs on the same individual

↳ its pair difference $D_i = X_{1i} - X_{2i}$

↳ $\text{Var}(D_i) = \text{Var}(X_{1i} - X_{2i}) = \sigma_1^2 + \sigma_2^2 - 2\text{Cov}(X_{1i}, X_{2i})$

$$\hookrightarrow CI \quad \bar{d} - t_{\alpha/2} \frac{sd}{\sqrt{n}} < \mu_D (= \mu_1 - \mu_2) < \bar{d} + t_{\alpha/2} \frac{sd}{\sqrt{n}}, \quad \text{where } \bar{d} = \bar{x}_1 - \bar{x}_2 = \sum d_i / n$$
$$sd = \sqrt{\frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n-1}}$$

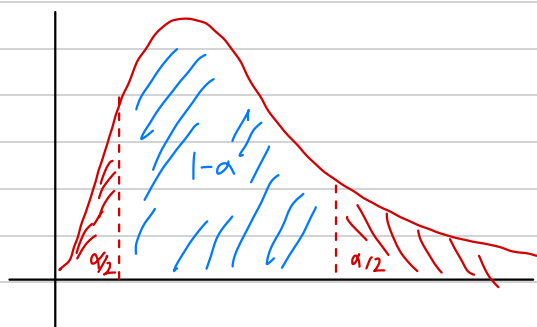
One sample, estimate variance

↳ Point estimate = S^2

↳ Define $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$

$$\hookrightarrow CI: \frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

↳ χ^2 table w/ $v=n-1$ d.o.f



Two sample estimate ratio of two variance

↳ Point estimate, $\hat{\sigma}_1^2/\hat{\sigma}_2^2$ der σ_1^2/σ_2^2

↳ Define $F = \frac{\sigma_1^2/\hat{\sigma}_1^2}{\sigma_2^2/\hat{\sigma}_2^2}$

↳ CI: $\frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} f_{\alpha/2}(v_2, v_1)$

Since $f_{1-\alpha/2}(v_1, v_2) = 1/f_{\alpha/2}(v_2, v_1)$ f-value w/ $v_1 = n_1 - 1$
 $v_2 = n_2 - 1$

Single Sample Proportion

↳ For $\hat{p}$ = proportion of success, $\mu_{\hat{p}} = p$, $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

CI: $\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$ Compact

$$\frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} - \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n}} < \frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} + \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n}}$$

more accurate,
Worse than Ep.

Sample size choice

$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2}$ exact

$n = \frac{z_{\alpha/2}^2}{4e^2}$ at least 100(1- α)% confidence, could be $\geq$

Two Sample Proportion

$$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} < p_1 - p_2 < (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$$

Maximum likelihood Estimation

Given indep. obs of $x_1, x_2, \dots, x_n$ from a p.d.f. (cont.) or p.m.f. (disc). $f(x; \theta)$

The maximum likelihood estimator $\hat{\theta}$ is that which maximizes the likelihood function:

$$L(x_1, x_2, \dots, x_n; \theta) = f(x; \theta) = f(x_1, \theta) \cdot f(x_2, \theta) \cdot \dots \cdot f(x_n, \theta)$$

Tip: Maybe take ln before differentiation

Steps:

(1) Write likelihood function

Ex: if $f(x; \mu) = \frac{e^{-\mu} \mu^x}{x!}$ (poisson)

$$\text{Then } L(x_1, x_2, \dots, x_n; \mu) = \frac{e^{-\mu} \mu^{x_1}}{x_1!} \cdot \frac{e^{-\mu} \mu^{x_2}}{x_2!} \cdot \dots = \frac{e^{-n\mu} \mu^{\sum x_i}}{\prod x_i!}$$

(2) "Optimize", take log

(3) Differentiate, find max.

Stats Chapter 10

Definitions

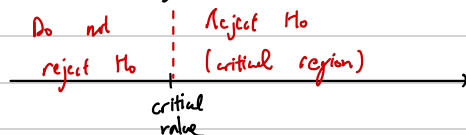
↳ A statistical hypothesis is an assertion or conjecture concerning one or more populations

↳ Null Hypothesis, H_0 (nullifies/opposes H_1)

↳ Alternative Hypothesis, H_1 (What we want to show)

- H_0 is maintained unless H_1 is supported by evidence "beyond a reasonable doubt"

↳ Test statistic, X



Errors

	Reality	
	H_0 true	H_0 false
Decision	Fail to reject H_0 Good Decision	Type II (miss) error
	Reject H_0 Type I error (false alarm)	Antispy

↳ Generally related, if $\alpha \downarrow$, $\beta \uparrow$ or vice versa

↳ Adjusting critical values $\rightarrow$ change critical region size

↳ Increasing sample size $n \rightarrow$ reduce α, β

↳ If H_0 false, β is max when true value of a parameter approaches the hypothesized value.

ie. The greater the distance between true value and hypothesized value, the smaller β is

α , Type I error

β , Type II error

$$\alpha = P(\text{reject } H_0 | H_0 \text{ true})$$

"the risk of incorrectly rejecting H_0 "

$$\beta = P(\text{not reject } H_0 | H_0 \text{ false})$$

$$= 1 - P(\text{reject } H_0 | H_0 \text{ false})$$

One / Two - tailed Tests

One-sided

$$H_0: \theta = \theta_0$$

$$H_1: \theta > \theta_0$$

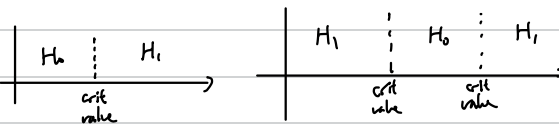
Two-sided

$$H_0: \theta = \theta_0$$

$$H_1: \theta \neq \theta_0$$

P-value

↳ The lowest level (of significance) at which the observed value of the test statistic is significant



Hypothesis Testing w/ fixed α

1. State H_0, H_1
2. Choose α fixed
3. Choose an appropriate test statistic and establish critical region based on α
4. Reject H_0 if statistic is in critical region.
Else, don't reject
5. Draw conclusion

Significance testing

1. State H_0, H_1
2. Choose test statistic
3. Compute P-value based on computed value of test statistic
4. Use judgement based on P-value and knowledge of scientific system.

One sample, known variance

$$\begin{aligned} H_0: \mu &= \mu_0, & z &= \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, & z > z_{\alpha/2} \text{ or } z < -z_{\alpha/2}, & -z_{\alpha/2} < z < z_{\alpha/2} \\ H_1: \mu &\neq \mu_0, & & & H_1, & H_0 \end{aligned}$$

↳ If $-z_{\alpha/2} < z < z_{\alpha/2} \rightarrow$ Don't reject H_0

↳ There will be a probability α of rejecting H_0 even if $\mu = \mu_0$

One sample, Unknown Variance

$$\begin{aligned} H_0: \mu &= \mu_0, & t &= \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, & t > t_{\alpha/2, n-1} \text{ or } t < -t_{\alpha/2, n-1}, & -t_{\alpha/2, n-1} < t < t_{\alpha/2, n-1} \\ H_1: \mu &\neq \mu_0, & & & H_1, & H_0 \end{aligned}$$

Two samples, known Variance

$$\begin{aligned} H_0: \mu_1 - \mu_2 &= d_0, & z &= \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}, & z > z_{\alpha/2} \text{ or } z < -z_{\alpha/2}, & -z_{\alpha/2} < z < z_{\alpha/2} \\ H_1: \mu_1 - \mu_2 &\neq d_0, & & & H_1, & H_0 \end{aligned}$$

Two samples, equal unknown Variances (pooled - t-test)

$$\begin{aligned} H_0: \mu_1 - \mu_2 &= d_0, & t &= \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}, & s_p^2 &= \frac{s_1^2(n_1-1) + s_2^2(n_2-1)}{n_1 + n_2 - 2}, & \text{reject } H_0 \text{ when } t > t_{\alpha/2, n_1+n_2-2} \text{ or } t < -t_{\alpha/2, n_1+n_2-2} \\ H_1: \mu_1 - \mu_2 &\neq d_0, & & & & & \end{aligned}$$

Two samples, unequal unknown variances

$$\begin{aligned} H_0: \mu_1 - \mu_2 &= d_0, & t' &= \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1'^2/n_1 + s_2'^2/n_2}}, & \text{d.o.f. } v &= \frac{(s_1'^2/n_1 + s_2'^2/n_2)^2}{(s_1'^2/n_1)^2/(n_1-1) + (s_2'^2/n_2)^2/(n_2-1)}, & \text{reject } H_0 \text{ when } t' > t_{\alpha/2, v} \text{ or } t' < -t_{\alpha/2, v} \\ H_1: \mu_1 - \mu_2 &\neq d_0, & & & & & \end{aligned}$$

Round down

Paired Observations

$$\begin{aligned} H_0: \mu_D &= d_0, & t &= \frac{\bar{d} - d_0}{s_d/\sqrt{n}}, & \bar{d} &= \sum \frac{d_i}{n} \\ H_1: \mu_D &\neq d_0, & & & \end{aligned}$$

★ one - sample problem

Sample size formulas

$$\delta = \mu - \mu_0$$

One-tailed, one sample

$$n \approx \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{\delta^2}$$

Two-tailed, one sample

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{\delta^2}$$

$$\delta = (\mu_1 - \mu_2) - d_0$$

One-tailed, Two samples

$$n \approx \frac{(z_{\alpha} + z_{\beta})^2 (\sigma_1^2 + \sigma_2^2)}{\delta^2}$$

Two-tailed, Two samples

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2 (\sigma_1^2 + \sigma_2^2)}{\delta^2}$$

$$= n_1 = n_2$$

$$= n_1 = n_2$$

Unknown variance

Estimate σ using σ'

Then using non-central t-distribution

① Find $\Delta = |\delta|/\sigma$

② Find corresponding n using table

One sample, single proportion test

- ① $H_0: p = p_0$
 $H_1 = \begin{cases} p < p_0 \\ p > p_0 \\ p \neq p_0 \end{cases}$
- ② Choose α
- ③ Test statistic, binomial variable X with $p = p_0$
- ④ Find x , number of successes, compute appropriate P-value
$$P = \begin{cases} P(X \leq x \text{ when } p = p_0) & (p < p_0) \\ P(X \geq x \text{ when } p = p_0) & (p > p_0) \\ P = 2P(X \leq x \text{ when } p = p_0) \text{ if } x < np_0 & (p \neq p_0) \\ P = 2P(X \geq x \text{ when } p = p_0) \text{ if } x > np_0 & (p \neq p_0) \end{cases}$$
- ⑤ Draw conclusion based on P-value, reject H_0 if $P \leq \alpha$

Large n , for $p_0 \approx 0$ or $p_0 \approx 1$

$$z = \frac{x - np_0}{\sqrt{np_0q_0}} = \frac{\hat{p} - p_0}{\sqrt{p_0q_0/n}} \quad \text{normal distribution, } \hat{p} = \frac{x}{n}$$

Two Samples, Two Proportions Test

- ① For large n , normally distributed
 $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2, \sigma_{\hat{p}_1 - \hat{p}_2}^2 = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$
- ② Pooled estimate of proportion $p, \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$
 $\hat{q} = 1 - \hat{p}$
- ③ Testing if $p_1 = p_2$:
$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}$$
- ④ Critical Region
$$\begin{aligned} z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2} & (p_1 \neq p_2) \\ z < -z_{\alpha} & (p_1 < p_2) \\ z > z_{\alpha} & (p_1 > p_2) \end{aligned}$$

One sample variance test

$H_0: \sigma^2 = \sigma_0^2$	Testing statistic:	Reject H_0 if:
$H_1: \begin{cases} \sigma^2 \neq \sigma_0^2 \\ \sigma^2 < \sigma_0^2 \\ \sigma^2 > \sigma_0^2 \end{cases}$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}, \nu = n-1$ σ_0^2 hypothesized variance	$\begin{cases} \chi^2 < \chi_{1-\alpha/2}^2 \text{ or } \chi^2 > \chi_{\alpha/2}^2 \\ \chi^2 < \chi_{1-\alpha}^2 \\ \chi^2 > \chi_{\alpha}^2 \end{cases}$

Two samples variance test

$H_0: \sigma_1^2 = \sigma_2^2$	Testing statistic:	Reject H_0 if:
$H_1: \begin{cases} \sigma_1^2 \neq \sigma_2^2 \\ \sigma_1^2 < \sigma_2^2 \\ \sigma_1^2 > \sigma_2^2 \end{cases}$	$f = \frac{s_1^2/s_2^2}, \nu_1 = n_1 - 1, \nu_2 = n_2 - 1$	$\begin{cases} f < f_{1-\alpha/2}(\nu_1, \nu_2) \text{ or } f > f_{\alpha/2}(\nu_1, \nu_2) \\ f < f_{1-\alpha}(\nu_1, \nu_2) \\ f > f_{\alpha}(\nu_1, \nu_2) \end{cases}$

Goodness of fit

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}, \quad v = k-1 \quad \longrightarrow \quad \begin{array}{ll} H_0: \text{Good fit} & \chi \leq \chi_{\alpha} \\ H_1: \text{Not Good fit} & \chi > \chi_{\alpha} \end{array}$$

o_i = observed frequency

e_i = expected frequency

(If $o_i < 5$, combine with adj cells)

Stats Chapter 11

Simple Linear Regression

↳ Relationship b/w a quantitative response y and a single explanatory variable x

$$y = \beta_0 + \beta_1 x + \epsilon$$

↳ β_0 → Constant
↳ β_1 → Independent var.
↳ ϵ → random error
↳ y → Dependent var.

Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

↳ Linear for coefficients

↳ $y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon$ is also linear cause $\beta_0, \beta_1, \beta_2$ linear

Random error

$$\epsilon \sim N(0, \sigma^2)$$

$$\hookrightarrow E(\epsilon) = 0$$

$$\hookrightarrow \text{Var}(\epsilon) = \sigma^2$$

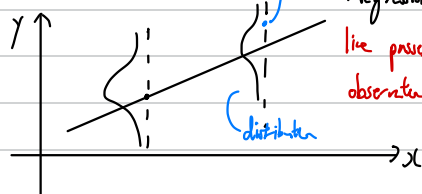
$$y = \beta_0 + \beta_1 x + \epsilon$$

$N(\beta_0 + \beta_1 x, \sigma^2) \sim N(0, \sigma^2)$

estimates estimators

$$b_0 \rightarrow \beta_0$$

$$b_1 \rightarrow \beta_1$$



Regression line, $\mu_{Y|X} = \beta_0 + \beta_1 x$
line passes through mean,
observations are on the distribution around mean

$$\hookrightarrow y_i \stackrel{iid}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Least squares fitting

Steps

① $SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$
sum of squares of error

② $\frac{\partial (SSE)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i)$

$\frac{\partial (SSE)}{\partial b_1} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_i) x_i$

③ Set SSE_{b_0} and $SSE_{b_1} = 0$

Normal equations:

$$n b_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i, \quad b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

④ Rearrange for b_0, b_1

$$b_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$b_0 = \frac{\sum_{i=1}^n y_i - b_1 \sum_{i=1}^n x_i}{n} = \bar{y} - b_1 \bar{x}$$

Properties of Least Squares Estimators

For $c_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}$, $i=1, 2, \dots, n \rightarrow \beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n c_i y_i$

Define $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$, $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$, $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
 $= \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n} = \sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n} = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n}$

①. X	$E(x)$	$Var(x)$	$N(\mu, \sigma^2)$
$\beta_1 = \frac{S_{xy}}{S_{xx}}$	β_1	$\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{S_{xx}}$	$\sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$
$\beta_0 = \bar{y} - \beta_1 \bar{x}$	β_0	$\left[\sum_{i=1}^n x_i^2 / \sum_{i=1}^n (x_i - \bar{x})^2 \right] \sigma^2 = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n S_{xx}}$	$\sim N(\beta_0, \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n S_{xx}})$

② $SSE = S_{yy} - 2b_1 S_{xy} + b_1^2 S_{xx}$
 $= S_{yy} - b_1 S_{xy}$

③ Unbiased estimate of $\sigma^2, s^2 = \frac{SSE}{n-2} = \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{n-2}$ ← *actual mke* *model mke* *mean squared error*
 $E(s^2) = \sigma^2$

Inference with Regression coefficients

Slope, β_1

→ CI:

$$b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}}, \quad v = n-2$$

→ Hypothesis testing:

$$H_0: \beta_1 = \beta_{10} \rightarrow t = \frac{b_1 - \beta_{10}}{s / \sqrt{S_{xx}}}$$

$$H_1: \beta_1 \neq \beta_{10}$$

Intercept, β_0

→ CI:

$$b_0 - t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2} < \beta_0 < b_0 + t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2}, \quad v = n-2$$

→ Hypothesis testing:

$$H_0: \beta_0 = \beta_{00} \rightarrow t = \frac{b_0 - \beta_{00}}{s \sqrt{\sum_{i=1}^n x_i^2 / n S_{xx}}}$$

$$H_1: \beta_0 \neq \beta_{00}$$

Coefficient of Determination, R^2

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Error sum of squares *Total corrected sum of squares*

Prediction

↳ Predict the mean response, $\mu_{Y|x_0}$ at $x=x_0$ using $\hat{y} = b_0 + b_1x$

Estimator: $\hat{Y}_0 = B_0 + B_1x_0$, $\hat{Y}_0 \sim N(E(\hat{Y}_0), \sigma_{\hat{Y}_0}^2)$

$$E(\hat{Y}_0) = \mu_{Y|x_0}$$

$$\sigma_{\hat{Y}_0}^2 = \sigma^2 \left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right]$$

LCI:

$$\hat{y}_0 - t_{\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} < \mu_{Y|x_0} < \hat{y}_0 + t_{\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \nu = n-2$$

↳ Prediction Interval (PI):

Given X , what is the range of values a single response y_0 can fall into

$$\hat{y}_0 - t_{\alpha/2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} < y_0 < \hat{y}_0 + t_{\alpha/2} s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \nu = n-2$$

Analysis of Variance (ANOVA)

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$SST = SSR + SSE$$

Regression sum of Squares, SSR → Variation in y -values explained by model

Error sum of Squares, SSE → Variation about the regression line

Hypothesis testing in ANOVA

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

$$f = \frac{SSR}{SSE/(n-2)} = \frac{SSR}{s^2}$$

→ Reject H_0 when $f > f_{\alpha}(1, n-2)$ [Only 2-sided]

Conclude a significant amount of variation in the response accounted for by the model

ANOVA Table

Variation	Sum of Squares	Dof, ν	Mean Square	f
Regression	SSR	1	SSR	$\frac{SSR}{s^2}$
Error	SSE	$n-2$	$s^2 = \frac{SSE}{n-2}$	
Total	SST	$n-1$		

↳ For testing $\beta_0 \neq 0$ in t -test,

$$t^2 = \frac{SSR}{s^2} = f$$

* A t -test for 0-slope two-sided = F -test

$$\text{Generally, } t^2(\nu) = f(1, \nu)$$

Multiple measurements

↳ n_i # of values for $x=x_i$

:

n_i # of values for $x=x_i$

$$\rightarrow \sum_{k=1}^i n_k = n \leftarrow \text{total obs}$$

↳ Then $y_{ij} = j^{\text{th}}$ value for r.v. Y_i

$$\bar{y}_i = \frac{\sum_{j=1}^{n_i} y_{ij}}{n_i}$$

$$\bar{y}_i = \bar{y}_i / n_i$$

Error sum of Squares

SSE = Lack-of-fit contribution + pure experimental error (PESS)

$$\textcircled{1} \text{ PESS} = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

and $s^2 = \text{PESS} / n - k$, unbiased estimate of σ^2

$\textcircled{2}$ Lack-of-fit sum of squares = SSE - PESS, $v = k - 2$

Complete ANOVA Table

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	f
Regression	SSR	1	SSR	$\frac{SSR}{s^2} \leftarrow \text{Test for } \beta_1 = 0 \text{ } H_0: \beta_1 = 0 / H_1: \beta_1 \neq 0$
Error	SSE	$n - 2$		
Lack of fit	SSE - PSSE	$k - 2$	$\frac{SSE - PSSE}{k - 2}$	$\frac{SSE - PSSE}{s^2(k - 2)} \leftarrow \text{Test for linearity (higher degree terms)}$
Pure Error	PSSE	$n - k$	$s^2 = \frac{PSSE}{n - k}$	$H_0: \text{Linear} / H_1: \text{Non-Linear}$
Tot	SST	$n - 1$		

Correlation

Population correlation coefficient: $\rho^2 = 1 - \frac{\sigma_y^2}{\sigma_y^2} = \beta_1^2 \frac{\sigma_x^2}{\sigma_y^2}$

Testing $\rho = 0$

↳ $-1 \leq \rho \leq 1$, $\rho = \pm 1$ only when $\sigma^2 = 0$

$$t = \frac{\hat{\rho} \sqrt{n-2}}{\sqrt{1-\hat{\rho}^2}}, \quad v = n-2 \quad \begin{matrix} H_0: \rho = 0 \\ H_1: \rho \neq 0 \end{matrix}$$

Sample estimate of ρ / Pearson correlation coefficient

$$\hat{\rho} = b_1 \frac{S_{xy}}{S_{xx}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

